1.67 Forces: Particles in Motion

Worked Solutions

1 a)

$F = 0.8 \times 1.6 = 1.28$ N

1 b)

$20 = 0.8m \Rightarrow m = 25$ kg

1 c)

$15 = 5a \Rightarrow a = 3$ ms^{-2}

1 d)

$F = 0.2 \times 0.8 = 0.16$ N

2 a)

$m = 5$

$$\text{R}(\uparrow): \quad Q - 5 \times 9.8 = 0 \Rightarrow Q = 49 \text{ N}$$

$$\text{R}(\rightarrow): \quad P - 10 = 5 \times 2 \Rightarrow P = 20 \text{ N}$$

2 b)

$m = 3$

$$\text{R}(\uparrow): \quad Q - 3 \times 9.8 = 0 \Rightarrow Q = 29.4 \text{ N}$$

$\text{R}(\rightarrow): \quad Q = 29$ N (2SF – AQA)

$$P - 8 = 3 \times 1.5 \Rightarrow P = 12.5 \text{ N}$$

2 c)

$m = 6.5$

$$\text{R}(\uparrow): \quad Q - 6.5 \times 9.8 = 0 \Rightarrow Q = 63.7 \text{ N}$$

$Q = 64$ N (2SF – AQA)

$$\text{R}(\rightarrow): \quad P - 40 = 6.5 \times -3 \Rightarrow P = 20.5 \text{ N}$$

3 a)

$m = 3$

$$\text{R}(\uparrow): \quad P - 3 \times 10 = 3 \times 2 \Rightarrow P = 36 \text{ N}$$

$P = 40$ N (1SF – AQA)

3 b)

$m = 7.5$

$$\text{R}(\uparrow): \quad P - 7.5 \times 10 = 7.5 \times 1.5 \Rightarrow P = 86.25 \text{ N}$$

$P = 90$ N (1SF – AQA)

3 c)

$m = 1.2$

$$\text{R}(\uparrow): \quad P - 1.2 \times 10 = 1.2 \times -0.8 \Rightarrow P = 11.04 \text{ N}$$

$P = 10$ N (1SF – AQA)

4 a)

$$\text{R}(\uparrow): \quad T - 2 \times 10 = 2 \times 3 \Rightarrow T = 26 \text{ N}$$

$T = 30$ N (1SF – AQA)

4 b)

$$\text{R}(\uparrow): \quad T - 2 \times 10 = 2 \times -1.5 \Rightarrow T = 17 \text{ N}$$

$T = 20$ N (1SF – AQA)

4 c)

$$\text{R}(\uparrow): \quad T - 2 \times 10 = 2 \times -2.5 \Rightarrow T = 15 \text{ N}$$

$T = 20$ N (1SF – AQA)

4 d)

$$\text{R}(\uparrow): \quad T - 2 \times 10 = 2 \times 0 \Rightarrow T = 20 \text{ N}$$

5)

$$\text{R}(\uparrow): \quad T - 0.5 \times 10 = 0.5 \times -0.2 \Rightarrow T = 4.9 \text{ N}$$

$T = 5$ N (1SF – AQA)

6)

$$\text{R}(\rightarrow): \quad 12 - R = 8 \times 0.1 \Rightarrow R = 11.2 \text{ N}$$

7)

$$\text{R}(\rightarrow): \quad 100 - R = m \times 0 \Rightarrow R = 100 \text{ N}$$

8 a)

$$\text{R}(\rightarrow): \quad 24 = m \times 1.5 \Rightarrow m = 16 \text{ kg}$$

$$\text{R}(\uparrow): \quad 16 \times 9.8 - w = 0 \Rightarrow w = 156.8 \text{ N}$$

$w = 160$ N (2SF – AQA)

8 b)

The box is a particle OR the box is rigid OR the force pushing the box is parallel to the floor

9)

$$\mathrm{R}(\rightarrow): \quad 2100 - 900 = 3000a \Rightarrow a = 0.4 \text{ ms}^{-2}$$

10)

$$\mathrm{R}(\downarrow): \quad 0.05 \times 9.8 - F = 0.05 \times 2.5$$

Total resistive force is 0.365 N

$F = 0.37$ N (2SF – AQA)

11 a)

$$0 = 18 + 5a \Rightarrow a = -3.6 \text{ ms}^{-2}$$

$$\mathrm{R}(\rightarrow): \quad F = 2000 \times -3.6 \Rightarrow F = -7200 \text{ N}$$

That is a breaking force of 7200 N

11 b)

There are no other external forces acting on the car. The car is modelled as a particle. The acceleration is uniform.

12)

$$0 = 20 + 6a \Rightarrow a = -\frac{10}{3} \text{ ms}^{-2}$$

$$\mathrm{R}(\rightarrow): \quad F - 450 = 900 \times -\frac{10}{3} \Rightarrow F = -2550 \text{ N}$$

That is a breaking force of 2550 N

13)

$$\mathrm{R}(\downarrow): \quad -2 + 0.4 \times 9.8 = 0.4 \times a \Rightarrow a = 4.8 \text{ ms}^{-2}$$

$$s = 0 + \frac{1}{2} \times 4.8 \times 1.5^2 = 5.4 \text{ m}$$

14 a)

$$20 = 9.8m \Rightarrow m = \frac{100}{49} = 2.04 \text{ kg (3 SF)}$$

$m = 2.0$ kg (2SF – AQA)

14 b)

$$\text{R}(\rightarrow): \quad 7 - 5.5 = \frac{100}{49} \times a \Rightarrow a = 0.735$$

$$v = 0 + 0.735 \times 10 = 7.35 \text{ ms}^{-1}$$

$v = 7.4$ ms^{-1} (2SF – AQA)

15)

$$\text{R}(\rightarrow): \quad 1.5 = 3 \times a \Rightarrow a = 0.5 \text{ ms}^{-2}$$

$$s = 0 + \frac{1}{2} \times 0.5 \times 4^2 = 4 \text{ metres}$$

16)

$$18^2 = 10^2 + 2 \times a \times 30 \Rightarrow a = \frac{56}{15} = 3.73 \text{ ms}^{-2}$$

$$\text{R}(\rightarrow): \quad F = 1650 \times \frac{56}{15} \Rightarrow F = 6160 \text{ N}$$

17)

$$\text{R}(\rightarrow): \quad -10000 = 2000 \times a \Rightarrow a = -5$$

$$u = 47 \text{ km h}^{-1} \Rightarrow \frac{235}{18} \text{ ms}^{-1}$$

$$0 = \left(\frac{235}{18}\right)^2 + 2 \times -5 \times s \Rightarrow s = 17.0 \text{ m}$$

So the car stops in time.

18 a)

$$25^2 = 0^2 + 2a \times 40 \Rightarrow a = 7.8125 \text{ ms}^{-2}$$

18 b)

$$\text{R}(\rightarrow): \quad F = 2250 \times 7.8125 \Rightarrow F = 17\,578.125 \text{ N}$$

$F = 17\,600$ N

This calculates the total force in the direction of motion so the assumption is that there is no resistance to motion such as friction from the road and air resistance. The van is also modelled as particle and is assumed to be rigid.

18 c)

$$\text{R}(\uparrow): \quad R - 2250 \times 9.8 = 0 \Rightarrow R = 22050 \text{ N}$$

$R = 22000$ (2SF – AQA)